

Mulugeta Solomon Abate

Software Engineer, Backend & System Design

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Profile

Software engineer specializing in backend and system design, shipping production services on AWS and PostgreSQL. I own REST APIs, asynchronous task pipelines, and multi-tenant platforms end to end — from schema design to deploy — and move across the full stack when the product needs it. M.Sc. Robotics, MEXT scholar.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL; working knowledge of Go, C++, C

Machine Learning & Evaluation: Model evaluation (mAP, IoU, F1, precision/recall), validation rules, data lineage; PyTorch, TensorFlow, YOLO, OpenCV, Pandas; ETL pipelines; agentic coding workflows and multi-agent orchestration

Backend & Full-Stack: FastAPI, NestJS, Node.js, Celery, SQLAlchemy, Prisma, REST APIs; React, Next.js, Tailwind CSS

Systems & Infrastructure: PostgreSQL, Redis, AWS (S3, RDS, Lambda, SQS, API Gateway, EC2), Docker, Linux, CI/CD; multi-tenancy, auth & RBAC, caching, rate limiting

Professional Experience

Sora Technology

Software Engineer

Oct 2024 – Present | Tokyo, Japan

- **Built the model-evaluation service** the detection models ship against — mAP, IoU, F1 and precision/recall with explicit definitions, lineage, and validation rules, making benchmarks reproducible and driving a **20% accuracy improvement**; drone-imagery breeding-site detection runs at **over 85% precision and recall**.
- **Designed and shipped the backend architecture for the SORA AI platform** — ingestion, transformation, and batch-upload services on AWS (S3, RDS/PostgreSQL, Lambda, SQS), cutting data iteration cycles **40%** and lifting upstream accuracy **30%**.
- **Built FloodSight**, a full-stack flood-risk platform (FastAPI + Celery + PostgreSQL + Next.js) — asynchronous task queues generate depth maps and downloadable reports (PNG, GeoTIFF, GeoJSON) for cross-functional flood analysis in Mozambique.
- **Made FloodSight's flood solver 12.5× faster** (0.663 s to 0.053 s per step, measured on staging) with byte-identical outputs, gated by per-step equality tests; a render-once cache cut repeat-load latency for reports and terrain overlays **15–30×**.
- **Lead developer on SORA's disease early-warning platform** (Ghana) — split authentication into its own Better Auth service with FastAPI and Next.js clients; delivery runs as review-gated work packages built with AI coding agents.
- **Implemented data-integrity and reconciliation checks** — automated coordinate validation between source GeoJSON and processed TIFF, catching discrepancies before downstream use and raising annotation accuracy by **25%**.
- **Automated preprocessing and query workflows** exposing clean, structured data to product and field stakeholders — eliminating **40+ hrs/month** of manual preparation.

Software / ML Intern

Jul 2024 – Oct 2024 | Tokyo, Japan

- **Migrated the detection service from YOLOv5 to YOLOv8**, benchmarking across multiple datasets for a **22% precision boost**.

Projects and Publications

fastapi-better-auth — open-source developer library

2026 – Present

- MIT-licensed **Python library** that verifies Better Auth sessions inside FastAPI, so Python services join a TypeScript-first auth stack instead of reimplementing it — shipped with **JWKS/JWT and end-to-end tests** against a real auth server in Docker, log-hygiene checks, a published compatibility matrix, and **release and upstream-drift canary CI**.

Karamu — Multi-tenant Restaurant SaaS (side project)

2025 – Present

- **Built solo, outside work** — a **multi-tenant SaaS** (NestJS + Prisma + PostgreSQL + Redis) with **role-based access control, rate limiting, and test-gated CI deploys to staging and production** — QR ordering, a real-time kitchen display, payments, and messaging.
- **Cut CPU cost per request to 0.45–0.73× of baseline** across the API — trimmed over-broad observability instrumentation, fixed a transaction-timeout path under load, and made connection-pool exhaustion fail fast as **503**; break points set by load test.

Education

Ritsumeikan University: CGPA 4.3/5.0 | Teaching Assistant (60+ graduate students)

Sep 2022 – Sep 2024 | Kyoto, Japan

M.Sc. in Robotics — AI, Advanced Robotics; co-authored an RSJ2024 paper on semantic line detection

Addis Ababa Science & Technology University: CGPA 3.81/4.00

Sep 2016 – Nov 2021 | Addis Ababa, Ethiopia

B.Sc. in Electromechanical Engineering — OOP, Database Systems, Data Structures, Network Security, Computer Vision

Africa to Silicon Valley (Backed by Google)

Feb 2024 – Feb 2025 | Addis Ababa, Ethiopia

Competitive Programming — 1,500+ hours (algorithms, data structures); 80+ alumni placed at Google, Bloomberg, Amazon